

- ❖ **Parasitoid:** Insect parasite of an arthropod which is parasitic during immature stages, but the adult is free living. Parasitoid is associated with one host and generally kills its host.
- ❖ **HISTORY**
 - In **1859**, V. M. Stern and colleagues formally proposed the concept.
 - They **developed** the ideas of economic damage, economic injury level, and economic threshold, collectively called the economic-injury level (EIL) concept. Although this concept was originally proposed in 1959, some of the ideas expressed had been discussed years earlier (1934) by W.D. Pierce.
- ❖ **Economic damage and damage Boundary**
 - **Economic damage** was originally defined as the amount of injury which will justify the cost of artificial control measures.
 - **Damage boundary**, defined as the lowest level of injury where damage can be measured

CATEGORIES OF PESTS

- ❖ **Key pest:** These are most severe and damaging pest. GEP lies well above the DB and EIL. E.g., Cotton boll worm, DBM, gram pod borer.
- ❖ **Major pest:** GEP is close to EIL but economically damage avoided by timely interventions. Sucking pests of cotton and rice.
- ❖ **Minor pest:** GEP lies below EIL and Damage boundary. Under favorable environmental condition the population may cross EIL and DB for a short interval and a single application of insecticides is usually enough to prevent damage. E.g., Thrips, mites and sugar cane mealy bug.
- ❖ **Regular pest:** Affect specific seasonal crops like cereals, pulses, fruits, passes through many generations during the crop period." E.g., rice stem borer
- ❖ **Sporadic pest:** Population of these insect is usually negligible but in certain year under favorable environmental condition they appear epidemic and crossing many times over EIL and DB. Required suitable cultural tactics to reduce population of these pests. White grub, hairy caterpillar, cut worm, grasshopper.
- ❖ **Potential pest:** These insects are presently not causing any economic damage. Therefore, as such should not be labeled as pests. The GEP lies below the DB and doesn't cross EIL even under favorable conditions. But any change in ecosystem may push their GEP higher and there is a danger of economic damage from these pests if control operations against the other categories of pests are undertaken in an indiscriminate manner. Spodoptera litura on cotton/Soybean. Army worm on wheat is example of potential pest.